An holographic perspective of confining field theories in type IIB

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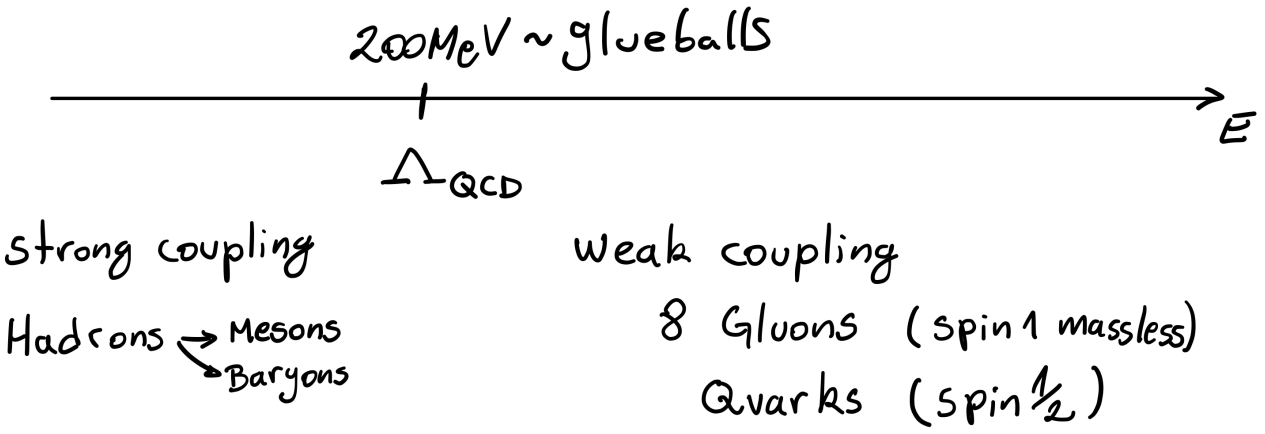
Outline

1. Introduction and AdS Solitons
2. Asymptotically locally flat configurations in $D = 4$
3. Uplift to 10 dimensions
4. Observables of the dual field theory
5. Flatten one of the 3-spheres
6. New black hole configurations

Introduction

Perturbation theory only works for small couplings.
Confinement is a strongly coupled phenomena.

QCD at low energies is a strongly coupled system

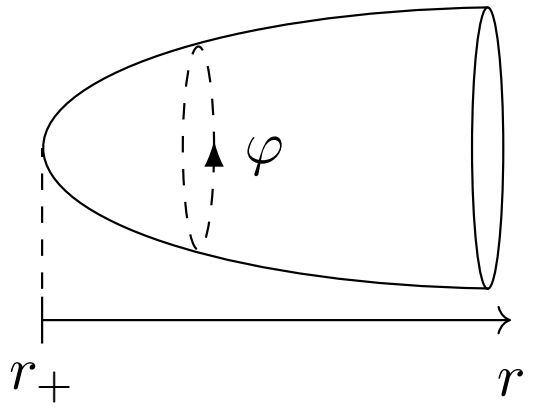


Holography allow us to have access to the strongly coupled dynamics doing perturbation theory in a dual system.

Introduction

- AdS/CFT duality relates a superconformal field theory with a string/M theory on spaces with AdS factor.
- After the conjectured duality, these ideas were extended to non-conformal situations [Itzhaki,Maldacena, Sonnenschein, Yankielowicz '98] [Witten 9803131] [Boonstra Skenderis Townsend, '98]
- Spaces with contractible cycle in the bulk(usually) leads to a confining dual theory.

$$ds^2 = \cdots + f(r)\, d\varphi^2$$
$$f(r_+) = 0$$



Solitons in AdS

Horowitz-Myers Soliton

Take planar Schwrazschild-AdS

$$ds^2 = -\left(\frac{r^2}{\ell^2} - \frac{r_+^3}{\ell^2 r}\right) dt^2 + \frac{dr^2}{\frac{r^2}{\ell^2} - \frac{r_+^3}{\ell^2 r}} + r^2 \left(dx^2 + dy^2\right) \ ,$$

Considering a double analytic continuation

$$t \to i\varphi \ , \qquad x \to it \ ,$$

We get Horowitz-Myers AdS soliton ['98 Horowitz,Myers]

$$\begin{aligned} ds^2 &= -r^2 dt^2 + \frac{dr^2}{\frac{r^2}{\ell^2} - \frac{r_+^3}{\ell^2 r}} + \frac{1}{\ell^2}\left(r^2 - \frac{r_+^3}{r}\right) d\varphi^2 + r^2 dy^2 \ , \\ L_\varphi &= \frac{4\pi\ell^2}{3r_+} \ . \qquad ds^2 \overset{r\to\infty}{\sim} r^2\left(-dt^2 + d\varphi^2 + dy^2\right) + \frac{dr^2}{r^2} \end{aligned}$$

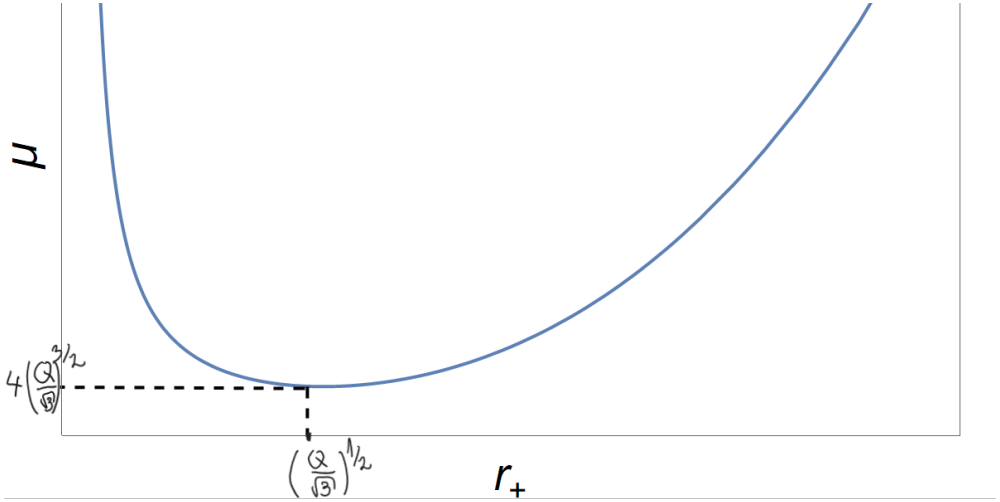
Anabalon-Ross Soliton

The lagrangian of $\mathcal{N}=2$ gauged supegravity in $D=4$ with susy variation

$$S=\frac{1}{2}\int d^4x\sqrt{-g}\left(R-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}+\frac{3}{2\ell^2}\right) \; + \; \text{fermions}$$

Charged planar Schwarzschild-AdS is solution

$$\begin{aligned} ds^2 &= -\left(\frac{r^2}{\ell^2}-\frac{\mu}{r}+\frac{Q^2}{r^2}\right)dt^2+\frac{dr^2}{\frac{r^2}{\ell^2}-\frac{\mu}{r}+\frac{Q^2}{r^2}}+r^2\left(dx^2+dy^2\right) \;, \\ A &= \left(\frac{2Q}{r}-\frac{2Q}{r_0}\right)dt \;. \end{aligned}$$



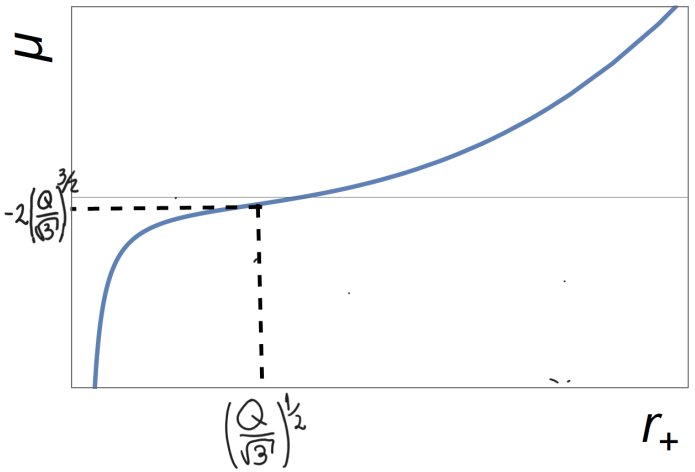
Anabalon-Ross Soliton

Consider the following analytic continuation

$$t \to ix \ , \qquad x \to i\varphi \ , \qquad Q \to -iQ \ .$$

leads to Anabalon-Ross soliton

$$\begin{aligned} ds^2 &= -r^2 dt^2 + \frac{dr^2}{\frac{r^2}{\ell^2} - \frac{\mu}{r} - \frac{Q^2}{r^2}} + \left(\frac{r^2}{\ell^2} - \frac{\mu}{r} - \frac{Q^2}{r^2}\right) d\varphi^2 + r^2 dy^2 \ , \\ A &= \left(\frac{2Q}{r} - \frac{2Q}{r_0}\right) d\varphi \ . \end{aligned}$$



Properties

- Uniparametric family of magnetic regular configuartion.
- Supersymmetric when $\mu = 0$.
- Asymptotically locally AdS.
- The energy from the vev stress tensor and the boundary current are

$$\langle T_{tt} \rangle = -\frac{\mu}{2\ell^2} \ , \qquad \langle J_\varphi \rangle = 2Q \ .$$

Asymptotically locally flat configurations in $\mathrm{D} = 4$

Freedman-Schwarz model

$\mathcal{N} = 4$ $SU(2) \times SU(2)$ gauged supergravity

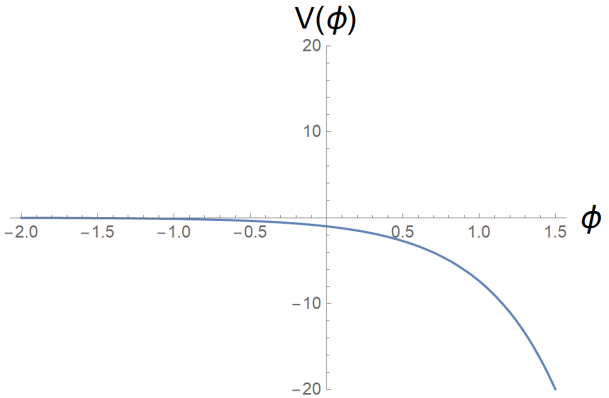
Field content

		Spin
$g_{\mu\nu}$	Metric	2
ϕ	Dilaton	0
a	Axion	0
A^i_μ, B^i_μ	two $su(2)$ gauge fields	1
ψ_μ	Gravitino	$3/2$
χ	Dilatino	$1/2$

Freedman-Schwarz model

$$
\begin{aligned}
S \;=\; & \int d^4x\sqrt{-g}\left[R-\frac{1}{2}\left(\partial\phi\right)^2+\frac{e_A^2+e_B^2}{8}e^{2\phi}-\frac{1}{2}e^{4\phi}\left(\partial \mathrm{a}\right)^2\right] \\
& -\frac{1}{2}\int \mathrm{tr}\left[e^{-2\phi}\left(F_A\wedge\star F_A+F_B\wedge\star F_B\right)+\mathrm{a}\left(F_A\wedge F_A+F_B\wedge F_B\right)\right] \\
& + \text{fermions } (\psi_\rho,\chi)
\end{aligned}
$$

We will focus on the $\mathrm{a}=0$ case regarding $F_A\wedge F_A+F_B\wedge F_B=0$.



Solutions of Freedman-Schwarz model

Electrovac ground state: $AdS_2 \times S^2$ and $AdS_2 \times \mathbb{R}^2$. [84' Freedman, Gibbons]

BPS topological black holes ['99 Klemm]

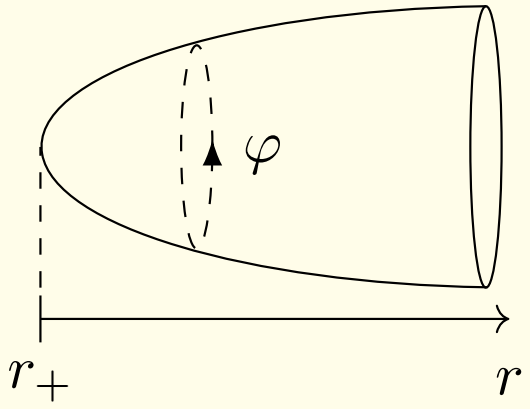
	M_4	BPS?
Magnetic field $\neq 0$	Spherical black holes	No
Magnetic field $= 0$	Planar black holes	Yes: naked singularities in general

Can we transformed the naked singularity BPS configuration into a regular configuration?

New D=4 regular background

In Einstein frame the metric of the soliton is '21 Canfora, Oliva, MO

$$
\begin{aligned}
ds_{4D}^2 &= -r^2 dt^2 + \frac{4dr^2}{f_s(r)} + r^2 f_s(r)\, d\varphi^2 + r^2 dx^2 \\
f_s(r) &= \frac{\left(e_A^2 + e_B^2\right)}{2} - \frac{m}{r^2} - \frac{2\left(Q_A^2 + Q_B^2\right)}{r^4}
\end{aligned}
$$



$$f_s(r_+) = 0 \implies \varphi \sim \varphi + \beta_\varphi$$

The configuration

$$
\begin{aligned}
\phi &= -\log r \\
A &= Q_A \zeta(r)\, d\varphi \qquad B = Q_B \zeta(r)\, d\varphi \\
\zeta(r) &= \left(r^{-2} - r_+^{-2}\right)
\end{aligned}
$$

- Regular magnetic soliton.

Properties of the 4D solution

- Asymptotically flat,

$$\lim_{r\to\infty} R^{\mu\nu}{}_{\rho\sigma} \to 0 \ ,$$

- BPS when $m = 0$ and $e_B Q_A \pm e_A Q_B = 0$. Preserves 4 killing spinors.

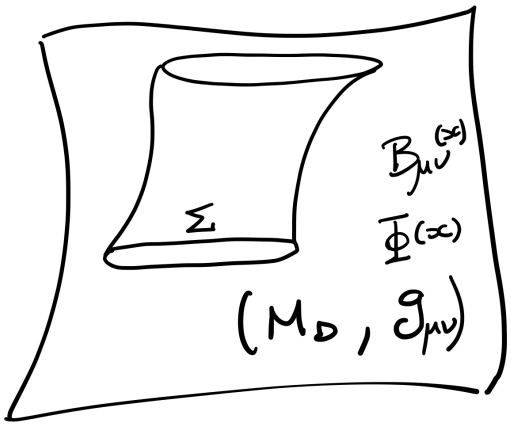
Killing spinors with $r^2 = \frac{2Q_A}{e_A}\cosh l$

$$\bar{\epsilon}_1 = e^{\frac{i\pi}{\beta}\varphi}(\cosh l)^{-1/4}\begin{pmatrix} -\sqrt{e_A^2+e_B^2}\sqrt{\cosh l - 1} \\ \sqrt{\cosh l + 1} \\ 0 \\ \frac{ie_A}{e_B}\sqrt{\cosh l + 1} \\ i\frac{\sqrt{e_A^2+e_B^2}}{e_B}\sqrt{\cosh l - 1} \\ -i\sqrt{\cosh l + 1} \\ 0 \\ \frac{e_A}{e_B}\sqrt{\cosh l + 1} \\ 0_{8\times 1} \end{pmatrix} , \quad \bar{\epsilon}_2,\ \bar{\epsilon}_3,\ \bar{\epsilon}_4 \ .$$

Uplift to 10 dimensions

Fast overview on strings

$$S_{Pol} = \frac{1}{2\pi\alpha'}\int_\Sigma [g_{\mu\nu}dX^\mu \wedge \star dX^\nu + B_{\mu\nu}dX^\mu \wedge dX^\nu + \dots$$



Conformal symmetry at quantum level

$$\beta = \mu\frac{\partial}{\partial\mu}\text{"couplings"} = 0 \ .$$

Implies relation between the background fields, in the limit $\alpha' \to 0$

$$S = \int d^Dx\sqrt{-g}e^{-2\Phi}\left(R + 4\left(\partial\Phi\right)^2 + \frac{1}{6}H_{\mu\nu\rho}H^{\mu\nu\rho} + \dots + \alpha'\left(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} + \dots\right)\right.$$

α' corrected black holes 2207.13214
Agurto,Chernicoff,Giribet,Oliva,MO

Type IIB

In the low energy limit $\alpha' \to 0$ of IIB string theory we get IIB supergravity

$$S_{\text{IIB}} = \int d^{10}x\sqrt{-g} \quad e^{-2\Phi}\left(R + 4\left(\partial\Phi\right)^2 - \frac{1}{2}\left|H_3\right|^2\right) - \frac{1}{2}\left|F_3\right|^2$$
$$-\frac{1}{2}\left|F_1\right|^2 - \frac{1}{4}\left|F_5\right|^2 - \frac{1}{2}\int C_4 \wedge H_3 \wedge F_3 + \text{ fermions}(\lambda, \psi_\mu)$$

where

$$F_1 = dC_0 \ , \qquad F_3 = dC_2 - C_0 H_3 \ , \qquad F_5 = dC_4 - C_2 \wedge H_3 \ .$$

In the $F_1 = F_5 = 0$ the susy transformations in the bosonic sector are

$$\delta\lambda = \frac{1}{2}\left(\partial\!\!\!/\Phi + \frac{1}{2}H\!\!\!/_3 \otimes \sigma_3 - \frac{1}{2}e^{\Phi}F\!\!\!/_3 \otimes \sigma_1\right)\epsilon$$
$$\delta\Psi_\mu dx^\mu = d\epsilon + \frac{1}{4}\left(\omega_{ab}\Gamma^{ab} + \frac{1}{2}H_{\mu ab}dx^\mu\Gamma^{ab}\sigma_3 + \frac{e^{\Phi}}{2}F\!\!\!/_3\Gamma_\mu dx^\mu\sigma_1\right)\epsilon$$

BPS check

To check whether a spacetime perserves some amount of supersymmetry we need to solve

$$\delta\lambda \equiv \mathbf{T}\epsilon = 0 \ , \qquad \delta\psi_\mu dx^\mu \equiv \mathbf{D}\epsilon = d\epsilon + \mathbf{W}\epsilon = 0 \ ,$$

for killing spinor ϵ. The strategy is

1. Compute the integrability condition for the Killing spinor

$$\mathbf{D} \wedge \mathbf{D}\epsilon = 0 \qquad \rightarrow \qquad \underbrace{(d\mathbf{W} + \mathbf{W} \wedge \mathbf{W})}_{\Xi_{\mu\nu}} \epsilon = 0$$

2. Solve the determiant equations

$$\det \Xi_{\mu\nu} = 0 \qquad \wedge \qquad \det \mathbf{T} = 0$$

   which imposes restrictions on the parameters of the background.

3. Finaly solve the linear equation

$$d\epsilon + \mathbf{W}\epsilon = 0$$

Kaluza-Klein on $S^3 \times S^3$

The Chamseddine-Volkov lift relates the NS sector of supergravity in 10 with the Freedman-Schwarz model

NS sector $(g_{10D}, \Phi, H_3) \xrightarrow{S^3 \times S^3} (g_{4D}, \phi, A_\mu, B_\mu, a)$ Freedman-Schwarz

The metric in 10D is the warpped (and fibred) product

$$g_{10D} \sim g_{4D} \times \left(S^3_{[\theta_A, \phi_A, \psi_A]} \times S^3_{[\theta_B, \phi_B, \psi_B]} \right)$$

Kaluza-Klein on $S^3 \times S^3$

Defining the Left invariant Maurer-Cartan 1-forms on each S^3

$$\begin{aligned}
\omega^1_A &= \cos\psi_A d\theta_A + \sin\psi_A \sin\theta_A d\phi_A \ , \\
\omega^2_A &= -\sin\psi_A d\theta_A + \cos\psi_A \sin\theta_A d\phi_A \ , \\
\omega^3_A &= d\psi_A + \cos\theta_A d\phi_A \ , \qquad \text{(same for } B\text{)} \ .
\end{aligned}$$

$$\begin{aligned}
ds^2\left(S^3_A\right) &= \frac{1}{4}\left[\left(\omega^1_A\right)^2 + \left(\omega^2_A\right)^2 + \left(\omega^3_A\right)^2\right] \\
&= \frac{1}{4}\left(d\theta^2_A + \sin^2\theta_A d\phi^2_A + \left(d\psi_A + \cos\theta_A d\phi_A\right)^2\right)
\end{aligned}$$

Kaluza-Klein on $S^3 \times S^3$

NS sector $(g_{10D}, \Phi, H_3) \overset{S^3\times S^3}{\longleftrightarrow} (g_{4D}, \phi, A_\mu, B_\mu, \mathrm{a} \to 0)$ Freedman-Schwarz

$$\Theta^i_A \equiv A^i - \frac{1}{e_A}\omega^i_A \ , \quad \Theta^i_B \equiv B^i - \frac{1}{e_B}\omega^i_B$$

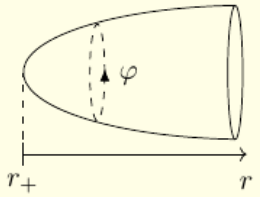
The KK reduction on $S^3 \times S^3$ + half susy braking conditions gives truncated to $a = 0$ plus the constraint $F_A \wedge F_A + F_B \wedge F_B = 0$

$$\begin{aligned} ds^2_{10D} &= e^{2\phi} ds^2_{4D} + 2\sum_i \left(\Theta^i_A \Theta^i_A + \Theta^i_B \Theta^i_B\right) \\ H_3 &= -2\sum_i \left(F^i_A \wedge \Theta^i_A + F^i_B \wedge \Theta^i_B\right) + 2e_A \Theta^1_A \wedge \Theta^2_A \wedge \Theta^3_A \\ &\quad +2e_B \Theta^1_B \wedge \Theta^2_B \wedge \Theta^3_B \\ \Phi &= \phi\left(x^m\right) \end{aligned}$$

New D=4 regular background

In Einstein frame the metric of the soliton is '21 Canfora, Oliva, MO

$$
\begin{aligned}
ds_{4D}^2 &= -r^2 dt^2 + \frac{4dr^2}{f_s(r)} + r^2 f_s(r)\, d\varphi^2 + r^2 dx^2 \\
f_s(r) &= \frac{\left(e_A^2 + e_B^2\right)}{2} - \frac{m}{r^2} - \frac{2\left(Q_A^2 + Q_B^2\right)}{r^4}
\end{aligned}
$$



$$f_s(r_+) = 0 \implies \varphi \sim \varphi + \beta_\varphi$$

The configuration

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A &= Q_A \zeta(r)\, d\varphi \qquad B = Q_B \zeta(r)\, d\varphi \\
\zeta(r) &= \left(r^{-2} - r_+^{-2}\right)
\end{aligned}
$$

- Regular magnetic soliton.

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Kaluza-Klein on $S^3 \times S^3$

New background in 10D

Lift of the soliton:NS background in IIB

In coordinates $[t, x, \varphi, r, \theta_A, \psi_A, \phi_A, \theta_B, \psi_B, \phi_B]$ the metric is

$$
\begin{aligned}
ds^2_{st} &= -dt^2 + dx^2 + f_s(r)\, d\varphi^2 + \frac{4dr^2}{r^2 f_s(r)} + \frac{2}{e_A}\left[\hat{\omega}_1^2 + \hat{\omega}_2^2 + (\hat{\omega}_3 - e_A Q_A \zeta(r)\, d\varphi)^2\right] \\
&\quad + \frac{2}{e_B}\left[\tilde{\omega}_1^2 + \tilde{\omega}_2^2 + (\tilde{\omega}_3 - e_B Q_B \zeta(r)\, d\varphi)^2\right] \ , \\
H_3 &= 2\zeta'(r)\, dr \wedge d\varphi \wedge \left(\frac{Q_A}{e_A}\hat{\omega}_3 + \frac{Q_B}{e_B}\tilde{\omega}_3\right) + \frac{2}{e_A^2}\hat{\omega}_1 \wedge \hat{\omega}_2 \wedge (e_A Q_A \zeta(r)\, d\varphi - \hat{\omega}_3) \\
&\quad + \frac{2}{e_B^2}\tilde{\omega}_1 \wedge \tilde{\omega}_2 \wedge (e_B Q_B \zeta(r)\, d\varphi - \tilde{\omega}_3) \ , \\
\Phi &= -\log r
\end{aligned}
$$

String coupling

$$g_s = e^{\Phi} = \frac{1}{r}$$

$\star H_3$–electric

	t	x	φ	r	θ_A	ϕ_A	ψ_A	θ_B	ϕ_B	ψ_B
NS5	×	×	×		×	×	×			
NS5	×	×	×					×	×	×

Lift of the soliton:NS background in IIB

Properties

- As $g_s = \frac{1}{r}$ it is well behaved for large r.
- It is an intersection of two stacks NS5 branes. It is dual to two intersecting little string theories.
- The dual theory becomes non-local in the UV.

To study a dual model trustable in the IR we will S-dualize.

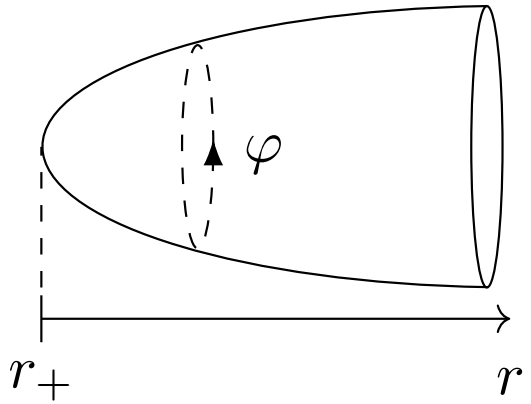
In Einstein frame in our background $(g_E, \Phi, H_3, F_3 = 0)$ S-duality is implemented as follows

$$\begin{aligned} g_E &\mapsto g_E \\ \Phi &\mapsto -\Phi \\ H_3 &\mapsto 0 \\ F_3 &\mapsto H_3 \end{aligned}$$

S-dual background: I-branes

In coordinates $(t, x, \varphi, r, \theta_A, \psi_A, \phi_A, \theta_B, \psi_B, \phi_B)$ the S-dual background with RR fields turned on

$$
\begin{aligned}
ds^2_{st} &= r\left\{-dt^2+dx^2+f_s(r)\,d\varphi^2+\frac{4dr^2}{r^2 f_s(r)}\right. \\
&\quad +\frac{N_B}{4}\left[\left(\omega^1_A\right)^2+\left(\omega^2_A\right)^2+\left(\omega^3_A-\sqrt{\frac{8}{N_B}}Q_A\zeta(r)\,d\varphi\right)^2\right] \\
&\quad \left.+\frac{N_A}{4}\left[\left(\omega^1_B\right)^2+\left(\omega^2_B\right)^2+\left(\omega^3_B-\sqrt{\frac{8}{N_A}}Q_B\zeta(r)\,d\varphi\right)^2\right]\right\}\,, \\
\Phi &= \log r\,, \qquad F_3 = \ldots \text{(same as before)}
\end{aligned}
$$

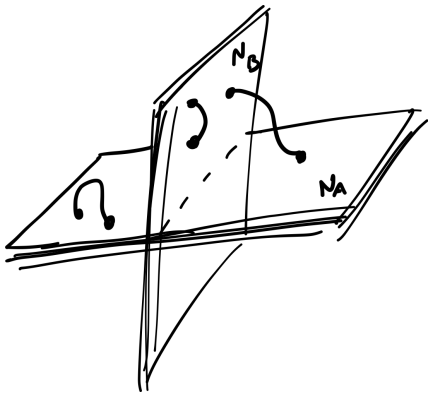


S-dual background: I-branes

2307.04183 Nunez, Stuardo, MO

In coordinates $(t, x, \varphi, r, \theta_A, \psi_A, \phi_A, \theta_B, \psi_B, \phi_B)$

$$
\begin{aligned}
F_3 \quad = \quad & 2N_B \mathrm{Vol}\left(S_A^3\right) + 2N_A \mathrm{Vol}\left(S_B^3\right) \\
& +2Q_A \sqrt{\frac{N_B}{8}} d\left(\zeta\left(r\right)\omega_A^3 \wedge d\varphi\right) + 2Q_B \sqrt{\frac{N_A}{8}} d\left(\zeta\left(r\right)\omega_B^3 \wedge d\varphi\right)
\end{aligned}
$$



	t	x	φ	r	θ_A	ϕ_A	ψ_A	θ_B	ϕ_B	ψ_B
D5$_A$	$\times$	$\times$	$\times$		$\times$	$\times$	$\times$			
D5$_B$	$\times$	$\times$	$\times$					$\times$	$\times$	$\times$

$$
N_A = \frac{1}{(2\pi)^2} \int_{S_B^3} F_3 \ , \quad N_B = \frac{1}{(2\pi)^2} \int_{S_A^3} F_3
$$

S-dual background: I-branes

The functions are

$$
\begin{aligned}
f_s(r) &= 1 - \frac{m}{r} - \frac{2\left(Q_A^2 + Q_B^2\right)}{r^4} = \frac{\left(r^2 - r_+^2\right)\left(r^2 - r_-^2\right)}{r^4} \\
r_\pm^2 &= \frac{1}{2}m \pm \frac{1}{2}\sqrt{m^2 + 8\left(Q_A^2 + Q_B^2\right)} \\
\zeta(r) &= \frac{1}{r^2} - \frac{1}{r_+^2}
\end{aligned}
$$

The curvature and string coupling are

$$
R = -\frac{3\left(3 - \frac{4N_A}{N_B} - \frac{4N_B}{N_A}\right)}{2r} + \frac{m}{r^3} - \frac{2\left(Q_A^2 + Q_B^2\right)}{r^5}\,, \qquad g_s = e^{\Phi} = r
$$

Properties of the I-brane configuration

- When $Q_A = Q_B = m = 0$ the background becomes singular: I-brane background

$$\mathcal{M}_{10} = \mathbb{R}^{1,2}_{t,x,\varphi} \times \mathbb{R}_r \times S^3 \times S^3$$

'06 Maldacena,Lin '06 Itzhaki,Kutasov,Seiberg

- At weak coupling the D5-D5 system preserves chiral supercharges and following the spectrum in the intersection contains chiral fermions which give rise to gauge anomalies, that is cured by inflow from the bulk of the D5 branes. '97 Green, Harvey, Moore
- At strong coupling in the field theory the system preserves $SO(2,1)$ with $SO(4) \times SO(4)$ R-symmetry and 16 supercharges. The 3D field theory has gauge group with Chern-Simons term

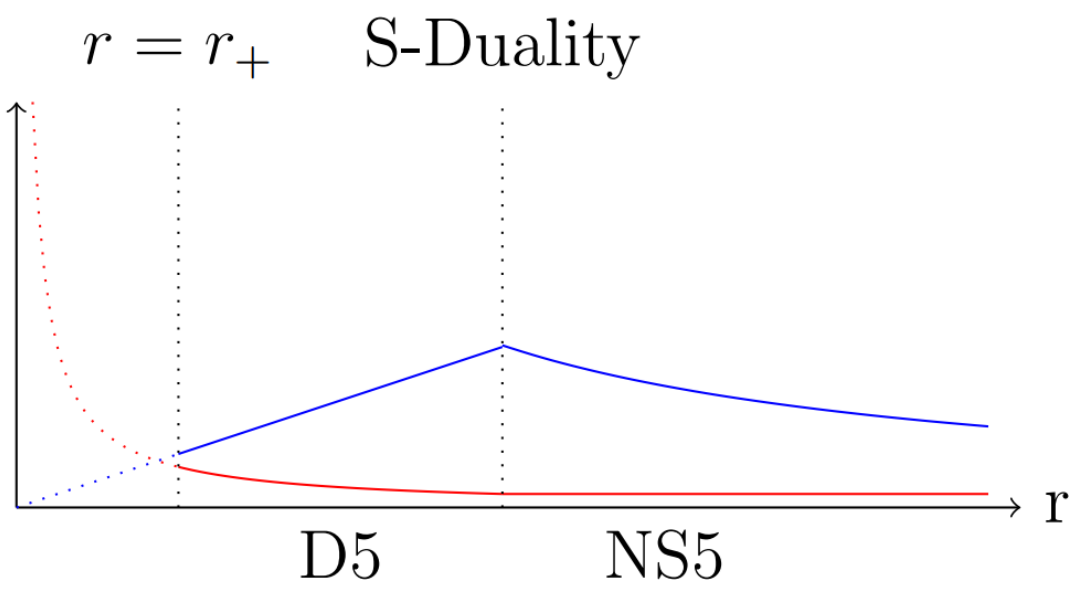
$SU(N_A)_{N_B} \times SU(N_B)_{N_A}$

Properties of the I-brane fibred configuration

- BPS when $m = 0$ and $N_B Q_A^2 = N_A Q_B^2$.
- Regular background for $r \geq r_+$.

The fibration ($Q_A \neq 0$,$Q_B \neq 0$) leads to a regular background.

- $e^{\Phi} = r$ grows at infinity $\Longrightarrow$ For large r better $S-$dualize.



Yang-Mills coupling

The action of a single Dp-brane is the **Dirac-Born-Infeld -Wess-Zumino** action

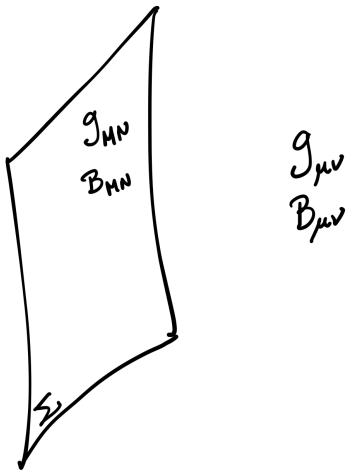
$$S_{Dp,DBI}\left[g_{MN},\mathcal{F}_{MN}\right] = T_p\int_\Sigma d^{p+1}x e^{-\Phi}\sqrt{-\det\left(g_{MN}+\mathcal{F}_{MN}\right)}\ ,$$

$$S_{Dp,WZ}\left[\mathcal{C},\mathcal{F}_{MN}\right] = -T_p\int_\Sigma \mathcal{C}\wedge e^{-\mathcal{F}_{MN}}\ ,$$

where

$$\mathcal{F}_{MN} = B_{MN} + 2\pi\alpha' F_{MN}\ ,$$

and $\mathcal{C}$ is a polyform.



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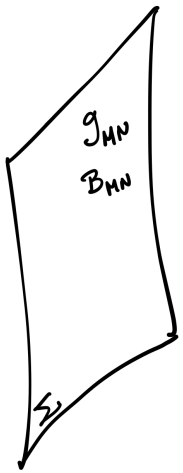


$g_{\mu\nu}$
$B_{\mu\nu}$

In the small field approximation

$$S_{Dp,DBI} = T_p\int_\Sigma d^{p+1}x e^{-\Phi}\sqrt{-\det g_{MN}}\left[1-\frac{1}{4}\left(2\pi\right)^2 F_{BC}F^{BC}+\mathcal{O}\left(F^3\right)\right]$$

Yang-Mills coupling

For our background, considering probe along $\Sigma_{(A)} = [t, x, \varphi, \theta_A, \phi_A, \psi_A]$. The Dirac-Born-Infeld action is

$$S^{(A)}_{D5,DBI} = T^{(A)}_{D5} L_\varphi \frac{32\sqrt{2}\pi^2}{e_A^3} r^2 \sqrt{f(r) + 2Q_B^2 \zeta(r)^2} \int dt dx \left(1 - \frac{(2\pi)^2}{4r^2} F_{\mu\nu} F^{\mu\nu}\right) ,$$

We read the Yang-Mills coupling from $\sim \frac{1}{g^2_{YM}} \int F_{\mu\nu} F^{\mu\nu}$ term

$$\frac{1}{g^2_{YM,A}} = 8\pi^4 T^{(A)}_{D5} N_B^{\frac{3}{2}} L_\varphi \sqrt{f(r) + 2Q_B^2 \zeta(r)^2} .$$

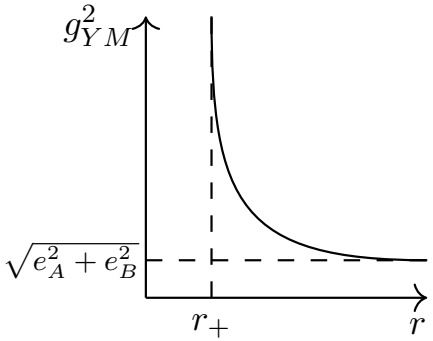
The dual field theory

- Two stacks of $D5$-branes intersecting at (t, x, φ)

$1+1$ effective field theory in the IR (t, x)

- **Yang-Mills coupling** $\mathcal{M}_5^{(A)} = (t, x, \varphi, \theta_A, \phi_A, \psi_A)$

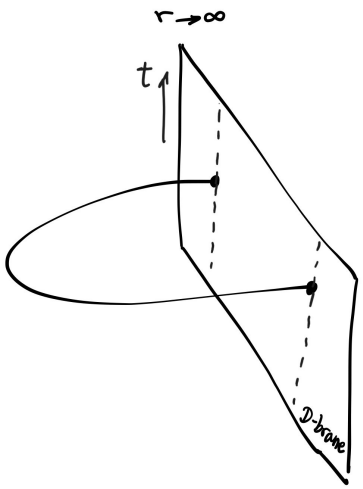
$$g_{YM,A}^{-2} \sim \sqrt{f(r) + 2Q_B^2 \zeta(r)^2}$$



Maldacena-Wilson Loop

The idea is to add a D-brane in the asymptotic region $r \to \infty$ and a macroscopic string attached to it, which dinamics is described by the Nambu-Goto action

$$S_{NG} = T_{F1} \int d^2\sigma \sqrt{-\det \boldsymbol{\gamma}} \ .$$



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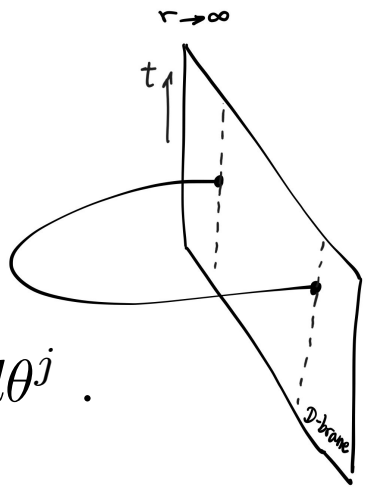


Considering a holographic background of the form

$$ds^2 = -g_{tt}dt^2 + g_{xx}d\vec{x}^2 + g_{rr}dr^2 + g_{ij}d\theta^i d\theta^j \ .$$

We assume that g_{tt}, g_{xx}, g_{rr} depends only on the radial coordinate. Consdering an embedding of the form

$$\begin{aligned} t &= \tau \ , \quad x = x(\sigma) \ , \quad r = r(\sigma) \ , \\ S_{NG} &= T_{F1} \int d\tau d\sigma \sqrt{g_{tt}(r)\, g_{xx}(r)\, x'^2 + g_{tt}(r)\, g_{rr}(r)\, r'^2} \end{aligned}$$

Maldacena-Wilson Loop

In this background the equations for the dynamics of the string moving in the background reduces to

$$\frac{dr}{d\sigma} = \pm \frac{dx}{d\sigma} V_{eff}\left(r\right) \ , \qquad V_{eff}\left(r\right) = \frac{F\left(r\right)}{CG\left(r\right)} \sqrt{F\left(r\right)^2 - C^2} \ ,$$

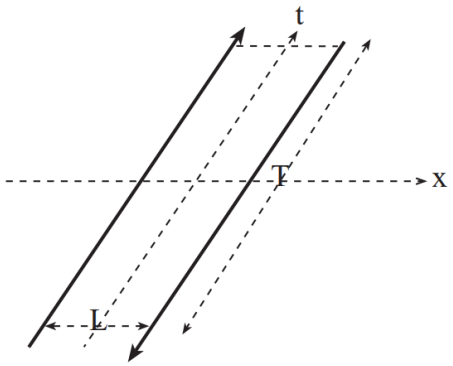
$F\left(r\right)^2 = g_{tt} g_{xx} \ , \quad G\left(r\right)^2 = g_{tt} g_{rr} \ .$

- In the simpler case we take $x\left(\sigma\right) = \sigma$.
- We set $C = F\left(r_0\right)$ where $r_0 \geq r_+$. Being r_0 the turning point of the string (U-shaped embeddings).

Maldacena-Wilson Loop

In the field theory the VEV of the Wilson loop operator can be approximated in the limit of infinite strip $T \to \infty$ as

$$\langle W(\mathcal{C})\rangle = \left\langle \frac{1}{N}\mathrm{tr}\ Pe^{i\oint_{\mathcal{C}} A}\right\rangle \sim e^{-TE(L)} \ ,$$



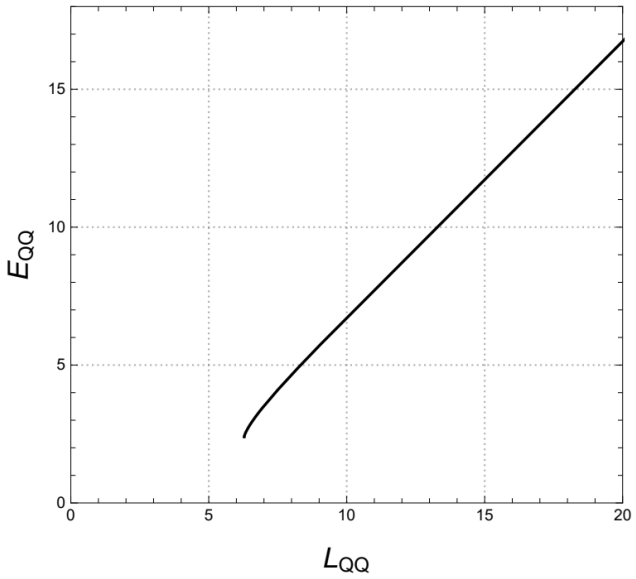
Sonnenschein hep-th/0003032

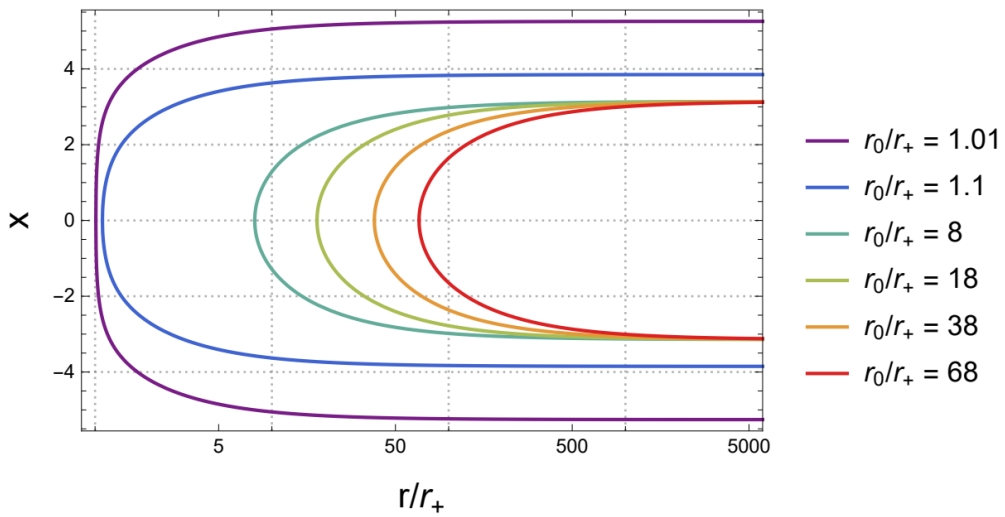
$E(L)$: quark-anti-quark potential.
A natural holographic candidate $\langle W(\mathcal{C})\rangle \sim e^{-S_{NG}}$.
Then the holographic expressions are

$$L_{QQ}(r_0) = 2\int_{r_0}^{+\infty} \frac{dz}{V_{eff}(z)} \ , \qquad F(r)^2 = g_{tt}g_{xx} \ , \ G(r)^2 = g_{tt}g_{rr}$$

$$E_{QQ}(r_0) = F(r_0)\,L_{QQ}(r_0) + 2\int_{r_0}^{+\infty} dz \frac{G(z)}{F(z)}\sqrt{F(z)^2 - F(r_0)}$$
$$-2\int_{r_+}^{+\infty} G(z)\,dz \ .$$

Maldacena-Wilson loop





- The concavity of the curve $E_{QQ}\left(L_{QQ}\right)$ is "downwards" $\Longrightarrow$ stable.
- For large separation L_{QQ} the energy grows linearly $\Longrightarrow$ confinement.
- There is a minimum for the separation, this scale is given by the Little String Theory scale.

Flatten one of the 3-spheres

Arxiv:2311.17998 in collaboration with Carlos Nuñez and Ricardo Stuardo

Background dual to 4+1 field theory

We will flatten one of the spheres

$$M_4 \times S^3 \times S^3 \to M_4 \times \mathbb{R}^3 \times S^3$$

then the dual theory is $4+1$ in the IR.

$$\begin{aligned} ds_{st}^2 &= r\left[-dt^2 + dx_4^2 + f_s(r)\, d\varphi^2 + \frac{N dr^2}{r^2 f_s(r)}\right. \\ &\qquad \left. + \frac{N}{4}\left(\omega_1^2 + \omega_2^2 + \left(\omega_3 - \sqrt{\frac{8}{N}} Q\zeta(r)\, d\varphi\right)^2\right)\right] \\ F_3 &= dC_2\ , \quad C_2 = \frac{N}{2}\psi \sin\theta d\theta \wedge d\phi + \sqrt{\frac{N}{2}} Q\zeta(r)\, \omega_3 \wedge d\varphi \\ \Phi &= \log r\ . \\ f_s(r) &= 1 - \frac{m}{r^2} - \frac{2Q^2}{r^4}\ , \quad \zeta(r) = \frac{1}{r^2} - \frac{1}{r_+^2}\ . \end{aligned}$$

	t	x_1	x_2	x_3	x_4	φ	r	θ	ϕ	ψ
D5	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$				

Glueballs

Spin 2 exicitacions in the field theory are necesarly boundstates. From the point of view of gravity it is enough to perturbate the metric and solve the equation for the perturbation consistently

$$g_{\mu\nu} \mapsto g_{\mu\nu} + \delta g_{\mu\nu} \ .$$

Find a consistent perturbation of the full system at linear level.

Glueballs

The type IIB equations in the $g_{\mu\nu} - \Phi - F_3$ sector in Einstein frame ["11 Bachas, Estes]

$$S_{IIB} = \frac{1}{2\kappa^2} \int d^{10}x\sqrt{-g} \left[R - \frac{1}{2} (\partial\Phi)^2 - \frac{1}{2!3!} e^{\Phi} F_{\mu\nu\rho} F^{\mu\nu\rho} \right] ,$$

are

$$G_{\mu\nu} - \frac{1}{2} \left(\partial_\mu \Phi \partial_\nu \Phi - \frac{1}{2} g_{\mu\nu} \partial_\rho \Phi \partial^\rho \Phi \right) - \frac{e^{\Phi}}{4} \left(F_{\mu\lambda\rho} F_{\nu}{}^{\lambda\rho} - \frac{1}{6} g_{\mu\nu} F_{\delta\rho\sigma} F^{\delta\rho\sigma} \right) = 0 ,$$

$$\Box\Phi - \frac{1}{12} e^{\Phi} F_{\delta\rho\sigma} F^{\delta\rho\sigma} = 0 , \qquad \nabla_\mu \left(e^{\Phi} F^{\mu\nu\rho} \right) = 0 .$$

Fact

The following perturbation is consistent

$$\delta g_{\mu\nu} = h_{\mu\nu} \neq 0 , \quad \delta F_{\mu\nu\rho} = 0 , \quad \delta\Phi = 0$$

in the traceless and transverse gauge

$$g^{\mu\nu} h_{\mu\nu} = 0 , \quad \nabla^\mu h_{\mu\rho} = 0 .$$

Glueballs

Let us consider a family of background metrics

$$
\begin{aligned}
g_{\mu\nu}dx^{\mu}dx^{\nu} &= e^{2A(\mathbf{y})}\left[\bar{g}_{ab}\left(\mathbf{x}\right)dx^{a}dx^{b}+\hat{g}_{ij}\left(\mathbf{y}\right)dy^{i}dy^{j}\right] \ , \\
F_3 &= \frac{1}{3!}f_{ijk}\left(\mathbf{y}\right)dy^{i}\wedge dy^{j}\wedge dy^{k} \ , \\
\Phi &= \Phi\left(\mathbf{y}\right) \ .
\end{aligned}
$$

$\bar{g}_{ab}$	$\{x^a\}$	spacetime of constant curvature k and dimension $p+1$
$\hat{g}_{ij}$	$\{y^i\}$	Inner space including the radial coordinate.

Let us consider a perturbation along the constant curvature space

$$
\delta g_{\mu\nu} = e^{2A(\mathbf{y})}\begin{pmatrix} h_{ab}\left(\mathbf{x}\right)\psi\left(\mathbf{y}\right) & 0 \\ 0 & 0 \end{pmatrix} \ .
$$

Glueballs

The resulting equation for the perturbation is

$$\psi \left(2k - \bar{\Box}_{\mathbf{x}}\right) h_{ab} - h_{ab}\hat{\Box}_{\mathbf{y}}\psi - 8h_{ab}\hat{g}^{ij}\partial_i A\partial_j\psi = 0 \ ,$$

Assuming that

$$\bar{\Box}_{\mathbf{x}} h_{ab} = \Lambda h_{ab} \ ,$$

The equation reduces to

$$\hat{\Box}_{\mathbf{y}}\psi + 8\hat{g}^{ij}\partial_i A\partial_j\psi + M^2\psi = 0 \ ,$$

where $M^2 = 2k - \Lambda$.

Glueballs for the background 4+1

Specializing for our background

e^{2A} $\bar{g}_{ab}(\mathbf{x})$

$$ds_E^2 = \sqrt{r}\left[-dt^2 + dx_4^2\right.$$

$$\left. + f_s(r)\, d\varphi^2 + \frac{N dr^2}{r^2 f_s(r)} + \frac{N}{4}\left(d\theta^2 + \sin^2\theta d\varphi^2 + \left(\omega_3 - \sqrt{\frac{8}{N}} Q\zeta(r)\, d\varphi\right)^2\right)\right]$$

Rescaling

$\hat{g}_{ij}(\mathbf{y})$

$$\psi(\mathrm{y}) = e^{-4A} f_s(r)^{-1/4} e^{i\frac{2\pi n}{L_\varphi}\varphi}\Theta(r)$$

and moving to the tortoise coordinate $r \to \rho \in [0, \infty[$ we get a Schrodinger like equation

$$-\frac{d^2\Theta}{d\rho^2} + \tilde{V}(\rho)\Theta = M^2\Theta\ ,$$

$$\tilde{V}(\rho) = \frac{1}{N} - \frac{1}{N\sinh^2\left(\frac{2\rho}{\sqrt{N}}\right)} + \frac{n^2}{N}\coth^2\left(\frac{\rho}{\sqrt{N}}\right)\left[1 + \lambda\tanh^2\left(\frac{\rho}{\sqrt{N}}\right)\right]^2$$

Glueballs for the background 4+1: Analitic solution

The Schrodinger equation admits an analitic solution. Considering $z = \cosh\left(\frac{\rho}{\sqrt{N}}\right)$. The radial profile of the metric perturbation which satisfy the boundary conditions

$$\delta g_{ab}|_{r_+} = 0 \ , \qquad \delta g_{ab}|_{r\to\infty} = 0 \ ,$$

is

$$\delta g_{ab} \sim e^{i\frac{2\pi n}{L\varphi}\varphi} \frac{z^{n\lambda}\left(z^2+\lambda\left(z^2-1\right)\right)^{\frac{1}{4}}}{\left(z^2-1\right)^{\frac{n}{2}}} \ {}_2F_1\left(-n-p, 1+p-n\lambda; 1-n\lambda; z^2\right)$$

the spectrum for the glueball masses reads

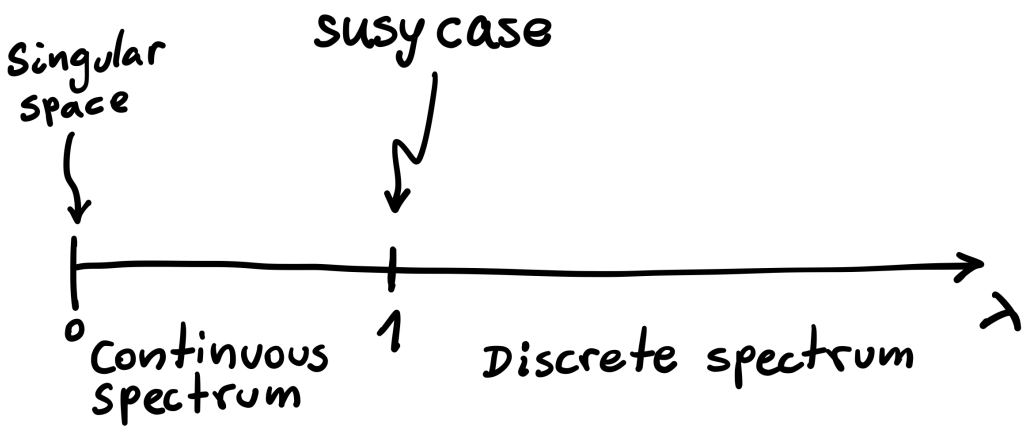
$$M^2 = \frac{1}{N}\left(1+n^2\left(\lambda+1\right)^2 - 4\left(p_{\max}-p\right)^2\right) \ ,$$

where

$$\begin{aligned} n &= 1, 2, \ldots \\ p &= 0, 1, 2, \ldots, \lfloor p_{\max} \rfloor \ , \qquad p_{\max} = \frac{1}{2}\left(n\left(\lambda+1\right)-1\right) \ . \end{aligned}$$

Glueballs for the background 4+1: Analitic solution

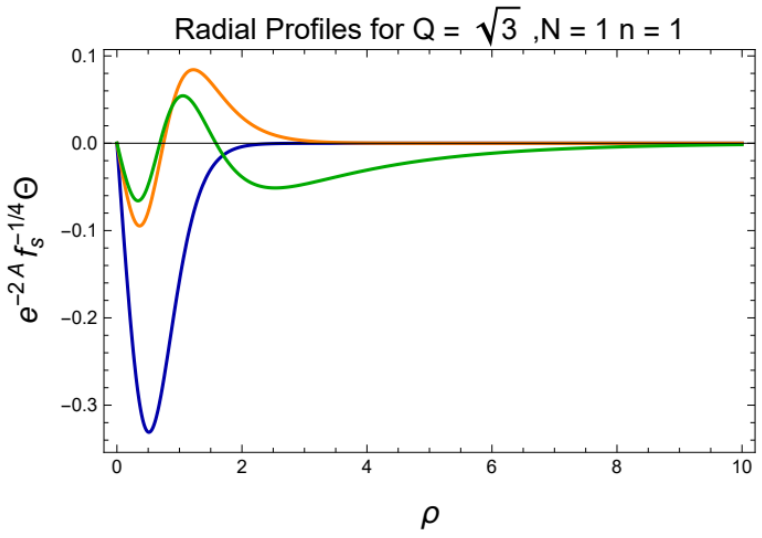
- The supersymmetric case $\lambda = 1$ separate the continuous spectrum $0 > \lambda > 1$ from the boundstates.

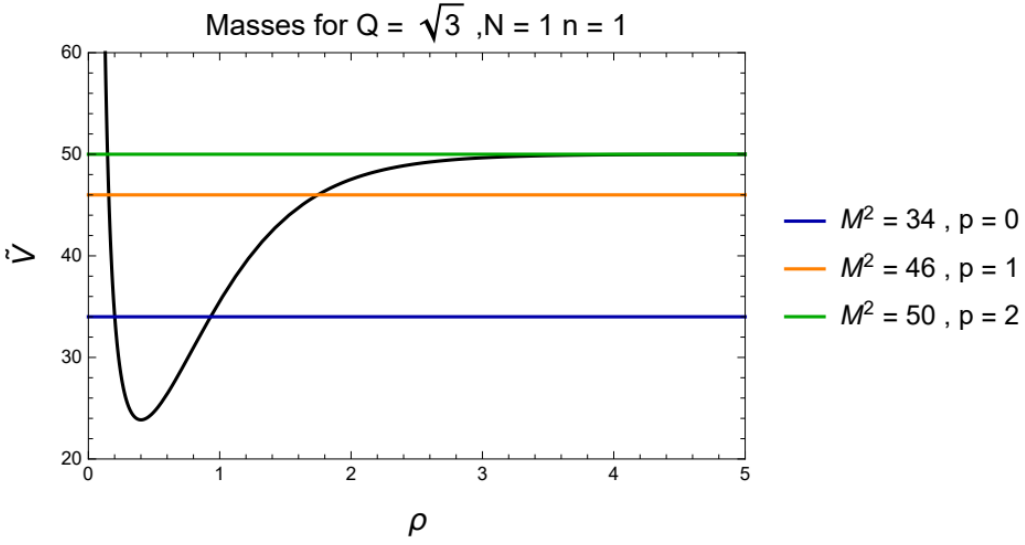


- The glueball states in the dual theory are $|n, p\rangle$

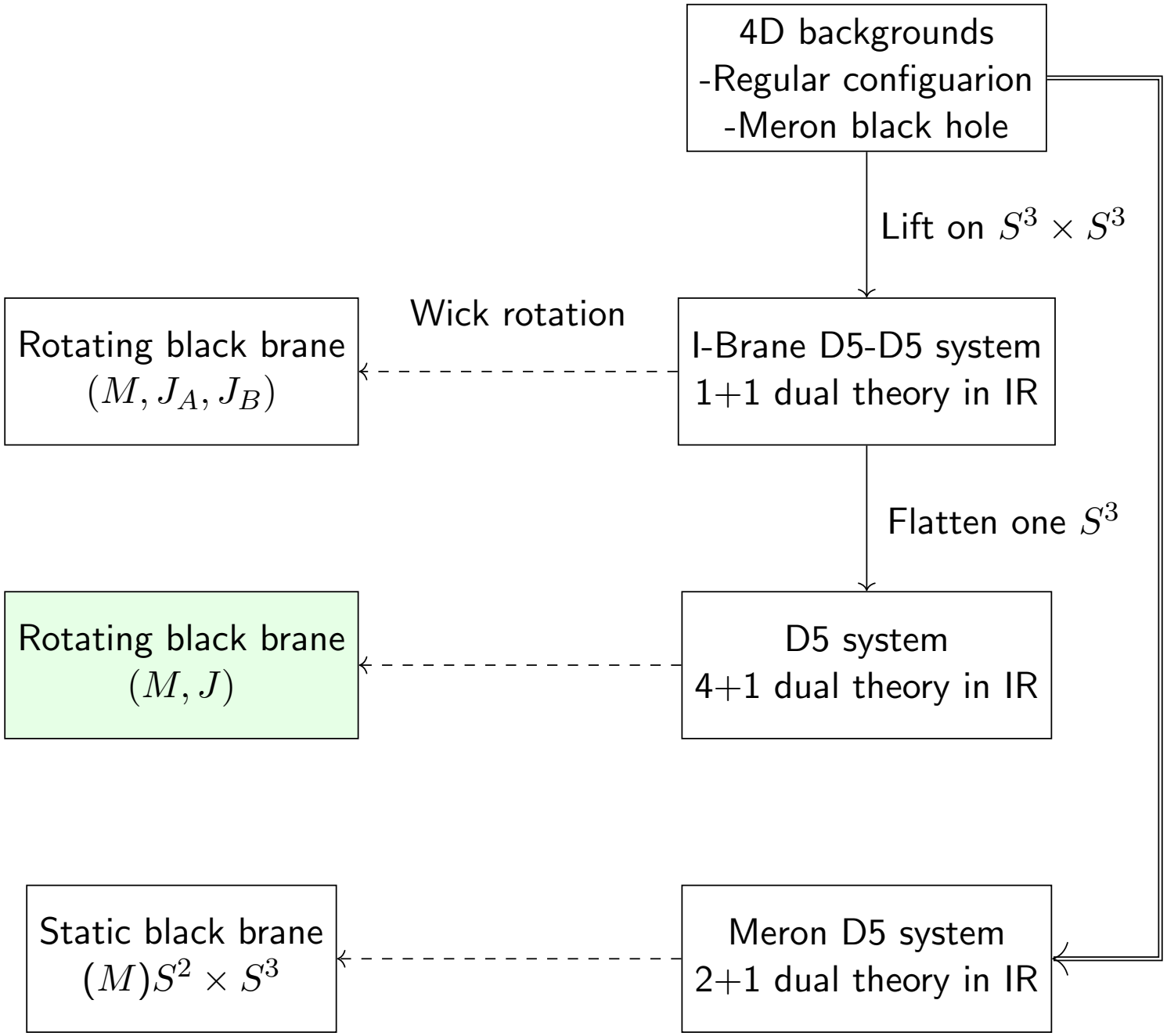
$$
\begin{aligned}
n &= 1, 2 \ldots \quad p = 0, 1 \ldots, \left\lfloor \frac{1}{2} \left(n \left(\lambda - 1 \right) - 1 \right) \right\rfloor \\
\omega^2 &= M_{n,p}^2 + \vec{k}^2
\end{aligned}
$$

Glueballs for the background 4+1: Analitic solution





Black holes configurations



$D = 4$
$\mathcal{N} = 4$
$SU(2) \times SU(2)$
gauged sugra

$D = 10$
IIB sugra

From the D5 background doing a double Wick rotationg we find the following solution of IIB sugra

$$ds_E^2 = \sqrt{r}\left[-dv\left(f(r)\,dv - 2\sqrt{N}r^{-1}dr\right) + dx_5^2\right]$$
$$+\frac{N\sqrt{r}}{4}\left[d\theta^2 + \sin^2\theta d\phi + \left(d\psi' + \cos\theta d\phi - \sqrt{8N^{-1}}Q\zeta(r)\,dv\right)^2\right]$$

$$\begin{aligned}
F_3 &= e^{-\frac{\Phi_0}{2}} d\left(\frac{N}{4}\cos\theta d\psi' \wedge d\phi + \sqrt{\frac{N}{2}}Q\zeta(r)\left(d\psi' + \cos\theta d\phi\right)\wedge dv\right) \\
\Phi &= \log r + \Phi_0 \\
f(r) &= 1 - \frac{m}{r^2} + \frac{2Q^2}{r^4}\,, \qquad r_\pm^2 = \frac{m}{2} \pm \frac{1}{2}\sqrt{m^2 - 8Q^2}
\end{aligned}$$

Iyer-Wald charges

We use Iyer-Wald formalism on

$$
\begin{aligned}
I_{\mathrm{IIB}} &= \frac{1}{2\kappa^2}\int_{\mathcal{M}} d^{10}x\sqrt{-g}\left(R-\frac{1}{2}\left(\partial\Phi\right)^2-\frac{1}{2}e^{\Phi}\left|F_3\right|^2\right) \\
&\quad + \int_{\partial\mathcal{M}} d^9x\sqrt{-h}\frac{1}{\kappa^2}\left(K+\mathcal{L}_{\mathrm{ct}}\right) \ .
\end{aligned}
$$

where n^μ is a normal outgoing vector to $\partial\mathcal{M}$

$$
\begin{aligned}
h_{\mu\nu} &= g_{\mu\nu}-n_\mu n_\nu \ , \\
K &= h^{\mu\nu}K_{\mu\nu} \ , \quad K_{\mu\nu}=h_\mu{}^\rho h_\nu{}^\sigma\nabla_\rho n_\sigma \ .
\end{aligned}
$$

and $\mathcal{L}_{\mathrm{ct}}$ is conterterm depending on the boundary fields. Different choices [Mann-Marolf 2006] [Kraus-Larsen-Siebelink 1999]

$$
\mathcal{L}_{\mathrm{ct}} \sim c_1\sqrt{\mathcal{R}} \ , \qquad \mathcal{L}_{\mathrm{ct}} \sim \frac{\mathcal{R}^{3/2}}{\left(\mathcal{R}^2-\mathcal{R}_{\mu\nu}\mathcal{R}^{\mu\nu}\right)^{1/2}}
$$

we used

$$
\mathcal{L}_{\mathrm{ct}} = -e^{-\frac{1}{4}\Phi} \ .
$$

Iyer-Wald in a nutshell

1. Compute a general variation of the (Bulk) action principle

$$\delta I = \int d^{10}x\sqrt{-g}\left[\mathcal{E}_{(\mathbf{f})}\cdot\delta\mathbf{f} + \nabla_\mu\Theta^\mu\left(\mathbf{f},\delta\mathbf{f}\right)\right]$$

2. Construct the Noether current associated to diffeomrofism generated by ξ^μ $\left(I_{\text{bulk}} = \int d^{10}x\mathcal{L}\right)$

$$J^\mu = \Theta^\mu\left(\mathbf{f},\delta\mathbf{f}\right) - \xi^\mu\mathcal{L}$$

3. Write the Noether current as a total derivative

$$J^\mu = \nabla_\nu q^{\mu\nu}\left(\mathbf{f},\xi\right)$$

For our case

$$\boldsymbol{Q}\left[\xi\right] = \star\boldsymbol{q} = -\frac{1}{\kappa^2}\left(\star d\xi + e^\Phi\xi\cdot C_2\wedge\star F_3\right)$$

Iyer-Wald charges

4. To compute the thermodynamics we consider the horizon generator

$$\boldsymbol{\xi} = \boldsymbol{t} + \Omega \boldsymbol{\psi} \qquad \text{s.t.} \qquad \boldsymbol{\xi}^2\big|_{\mathcal{H}_+} = 0 \ .$$

then the charges are

$$\begin{aligned} M[\boldsymbol{t}] &= \int_\infty \boldsymbol{Q}[\boldsymbol{t}] - \boldsymbol{t}\cdot\boldsymbol{B} \\ J[\boldsymbol{\psi}] &= -\int_\infty \boldsymbol{Q}[\boldsymbol{\psi}] \\ S &= \frac{1}{T}\int_{\mathcal{H}_+} \boldsymbol{Q}[\boldsymbol{\xi}] \\ T &= \frac{\kappa_s}{2\pi}\ , \qquad \xi^\mu \nabla_\mu \xi^\nu = \kappa_s \xi^\nu\big|_{\mathcal{H}_+} \end{aligned}$$

D5 black brane charges

Considering the vectors

$$\boldsymbol{t} = \partial_v \ , \qquad \boldsymbol{\psi} = \frac{\partial}{\partial \psi} \ , \qquad \Omega = \sqrt{\frac{8}{N}} \frac{Q}{r_+^2}$$

$\boldsymbol{\xi} = \boldsymbol{t} + \Omega \boldsymbol{\psi}$. They give the following charges $\Phi_0 = \log \frac{N^2}{16}$

$$\begin{aligned}
M &= \frac{N\left(r_+^4 + 2Q^2\right)}{\kappa^2 r_+^2} L_{x_1} L_{x_2} L_{x_3} L_{x_4} L_{x_5} \pi^3 \ , \\
J &= \frac{\sqrt{2N^3} Q}{\kappa^2} L_{x_1} L_{x_2} L_{x_3} L_{x_4} L_{x_5} \pi^3 \ , \\
S &= \frac{4N^{3/2} r_+^2 \left(r_+^4 + 2Q^2\right)}{\left(r_+^4 - 2Q^2\right)} L_{x_1} L_{x_2} L_{x_3} L_{x_4} L_{x_5} \pi^4 \ , \\
T &= \frac{r_+^4 - 2Q^2}{2\sqrt{N} \pi r_+^4} \ .
\end{aligned}$$

Note: If we insist in consider $S = \mathrm{Area}/4$ then $dM = TdS - 2\Omega dJ$.

Summary and outlook

- Study the black brane solution constructed from the double Wick rotation of the background. Can we compute the entropy of the black brane from a $2+1$ field theory compactified on a torus?
- Study the string $\sigma-$model in the full solution. Beyond the particular case with $Q_A = Q_B = 0$ that we revisited.
- It would be of interest to generalize the metric and fluxes of the background by adding warp factor in from the Maurer-Cartan 1-forms with a more general fibration.

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Muchas gracias!